Introduction

In Chapter 3, you learned how to fine-tune a model for **text classification**.
In this chapter, we will explore several important language tasks that are essential for both traditional NLP and modern LLMs.

What you will learn in this chapter

- Token classification
- Masked language modeling (like BERT)
- Summarization (shortening text)
- Translation (converting one language to another)
- Causal language modeling pretraining (like GPT-2)
- Question answering

Why are these important?

All of these tasks form the **foundation of how Large Language Models (LLMs) work**.

If you understand these topics well, you will be able to:

- Understand modern AI models more easily
- Use LLMs in your own projects
- Perform better in research or Kaggle competitions

What you need to know before this chapter

To understand this chapter properly, you should know concepts from previous chapters:

- Chapter 3 → Trainer API and Accelerate library
- Chapter 5 → Datasets library
- Chapter 6 → Tokenizers library

Also, like in Chapter 4, we will upload our results to the **Model Hub**.

This means that in this chapter, everything you learned earlier will come together.

Structure of the chapter

Each section of this chapter can be read independently.
In each section, you will learn how to:

- Train a model using the Trainer API
  or
- Train a model using your own custom training loop with Accelerate

Trainer API vs Custom Training Loop

Trainer API

- Very easy to use
- Good for fine-tuning or training models
- You don't need to worry about internal details

Accelerate (Custom Training Loop)

- Gives more control
- Easier to customize
- Better for advanced use cases

Final Note

You can choose to:

- Focus only on the Trainer API
  or
- Focus only on the custom training loop with Accelerate

You should focus on the part that matches your learning goal.